Prob. 9. (2aT. Ch. 7. § 8.6)

This is a practice to get nonzero terms in the F.S. No need to derive F.S. for (8.32) – (8.33). For all such problems, you can check by writing out the terms of your resulting series to see if you recover the nonzero terms of your original series.

(8.18) Show that

(1) $$\frac{\pi}{2} + \sum_{n=1}^{\infty} 2 \cdot \frac{(-1)^n - 1}{\pi n^2} \cdot \cos(nx) = \frac{\pi}{2} - \frac{4}{\pi} \sum_{k=1}^{\infty} \frac{\cos[(2k-1)x]}{(2k-1)^2}$$

(2) $$RHS = \frac{\pi}{2} - \frac{4}{\pi} \cdot \sum_{k=1}^{\infty} \frac{\cos[(2k-1)x]}{(2k-1)^2}$$

(3)

$k = 1.$

$$\frac{\cos[(2\cdot 1 - 1)x]}{(2\cdot 1 - 1)^2} = \frac{\cos(x)}{1}$$

$k = 2$

$$\frac{\cos[(2\cdot 2 - 1)x]}{(2\cdot 2 - 1)^2} = \frac{\cos(3x)}{9}$$

$k = 3$

$$= \frac{\cos(5x)}{25}$$

$k = 4$

$$= \frac{\cos(7x)}{49}, \quad \text{So}$$

...

(4) $$RHS = \frac{\pi}{2} - \frac{4}{\pi} \cdot \left[\frac{\cos(x)}{1} + \frac{\cos(3x)}{9} + \frac{\cos(5x)}{25} + \frac{\cos(7x)}{49} + \dots \right]$$

$$= \frac{\pi}{2} - \frac{4\cos(x)}{\pi} - \frac{4 \cdot \cos(3x)}{9\pi} - \frac{4\cos(5x)}{25\pi} - \frac{4\cos(7x)}{49\pi} \dots$$

Whereas, LHS,

(5) $$LHS = \frac{\pi}{2} + \sum_{n=1}^{\infty} 2 \cdot \frac{(-1)^n - 1}{\pi n^2} \cdot \cos(nx) =$$

$n = 1$

$$2 \cdot \frac{(-1)^1 - 1}{\pi \cdot (1)^2} \cdot \cos(1x) = 2 \cdot \frac{(-2)}{\pi} \cos(x) = \frac{-4}{\pi} \cos(x)$$

$n = 2$

$$2 \cdot \frac{(-1)^2 - 1}{\pi \cdot (2)^2} \cos(2x) = 2 \cdot \frac{0}{4\pi} \cos(2x) = 0$$

$n = 3$

$$2 \cdot \frac{(-1)^3 - 1}{\pi \cdot (3)^2} \cos(3x) = 2 \cdot \frac{(-2)}{\pi \cdot 9} \cos(3x) = \frac{-4 \cos(3x)}{9\pi}$$

(6) $n = 4$

$$2 \cdot \frac{(-1)^4 - 1}{\pi \cdot (4)^2} \cos(4x) = 0$$

$n = 5$

$$2 \cdot \frac{(-1)^5 - 1}{\pi \cdot (5)^2} \cdot \cos(5x) = \frac{-4 \cos(5x)}{25\pi}$$

$n = 6$

$= 0,$

$n = 7$

$$2 \cdot \frac{(-1)^7 - 1}{\pi \cdot (7)^2} \cos(7x) = \frac{-4 \cos(7x)}{49\pi}$$

(7) $$LHS = \frac{\pi}{2} + \frac{4}{\pi} \cos(x) - \frac{4 \cos(3x)}{9\pi} - \frac{4 \cos(5x)}{25\pi} - \frac{4 \cos(7x)}{49\pi} - \ldots$$

Of course, $\overset{(7)}{LHS} \equiv \overset{(4)}{RHS}$, by the results of (4) & (7), verifying that we can recover the RHS out of LHS.

As we've noticed in process (6), $n =$ <u>even</u> values cancel out in the sum, leaving us to add all <u>odd</u> contributions. So,

(8) Let $\boxed{n = 2k - 1}$, then LHS in (5) is

(9) $$LHS = \frac{\pi}{2} + \sum_{k=1}^{\infty} 2 \cdot \frac{(-1)^{2k-1} - 1}{\pi (2k-1)^2} \cdot \cos[(2k-1)x] =$$

$$= \frac{\pi}{2} + \sum_{k=1}^{\infty} \frac{2 \cdot (-2)}{\pi (2k-1)^2} \cdot \cos[(2k-1)x] =$$

$$= \frac{\pi}{2} + \sum_{k=1}^{\infty} \frac{-4}{\pi} \cdot \frac{\cos[(2k-1)x]}{(2k-1)^2} =$$

$$= \frac{\pi}{2} - \frac{4}{\pi} \sum_{k=1}^{\infty} \frac{\cos[(2k-1)x]}{(2k-1)^2} \equiv RHS. \checkmark$$

(For the rest of the probs, I try to do the calculation processes shorter, especially, maybe skipping (3) & (6).

(8,31) $\frac{2}{\pi} \cdot \sum_{n=1}^{\infty} \frac{1-(-1)^n}{n} \sin(nx)$

(1)

(2) $n = 1$ $\frac{1-(-1)^1}{1} \sin(1 \cdot x) = \frac{2}{2} \sin x$ = ~~non zero~~

(3) $n = 2$ $\frac{1-(-1)^2}{1} \sin(2 \cdot x) = \frac{0}{1} \sin(2x) = 0$

So let

$n = 2k-1$, then given series becomes

$$\frac{2}{\pi} \sum_{k=1}^{\infty} \frac{1-(-1)^{2k-1}}{(2k-1)} \cdot \sin[(2k-1)x] =$$

$$= \frac{2}{\pi} \cdot \sum_{k=1}^{\infty} \frac{2}{(2k-1)} \sin[(2k-1)x] =$$

$$= \frac{4}{\pi} \cdot \sum_{k=1}^{\infty} \frac{\sin[(2k-1)x]}{(2k-1)}.$$

Hence

$$\frac{2}{\pi} \cdot \sum_{n=1}^{\infty} \frac{1-(-1)^n}{n} \sin(nx) = \frac{4}{\pi} \sum_{k=1}^{\infty} \frac{\sin[(2k-1)x]}{(2k-1)}$$

(8.32). Same directions.

(1) $$\underbrace{x(\pi - x)}_{LHS} = \underbrace{\frac{\pi^2}{6} - 2\cdot\sum_{n=1}^{\infty} \frac{1+(-1)^n}{n^2}\cos(nx)}_{RHS.}$$

let

(2) $$RHS = \frac{\pi^2}{6} - 2\cdot\sum_{n=1}^{\infty} \frac{1+(-1)^n}{n^2}\cos(nx).$$

(3) $\boxed{n = 2k}$, $\underline{k = 1, 2, \ldots}$ then

$$\frac{1+(-1)^{2k}}{(2k)^2}\cos(2kx) = \frac{2}{4k^2}\cos(2kx) =$$

$$= \frac{1}{2k^2}\cos(2kx).$$ while

(4) $\boxed{n = 2k-1}$ $k = 1, 2, \ldots$, then

$$\frac{1+(-1)^{2k-1}}{(2k-1)^2}\cdot\cos[(2k-1)x] = \frac{0}{(2k-1)^2}\cdot\cos(\ldots) = 0$$

So $n = 2k-1$ (odd) terms are zero, leaving us only even terms $\boxed{n = 2k}$ to add in the RHS.

(5) $$RHS = \frac{\pi^2}{6} - 2\cdot\sum_{k=1}^{\infty} \frac{1+(-1)^{2k}}{(2k)^2}\cos(2kx) =$$

$$= \frac{\pi^2}{6} - 2\cdot\sum_{k=1}^{\infty} \frac{2}{4k^2}\cos(2kx) = \frac{\pi^2}{6} - \sum_{k=1}^{\infty} \frac{\cos(2kx)}{k^2}. \Rightarrow$$

(6) $$x(\pi - x) = \frac{\pi^2}{6} - \sum_{k=1}^{\infty} \frac{\cos(2kx)}{k^2}.$$ (wierd Σ of even f-ns = neither f-n)?

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